

Diagram Development Methodology

The diagrams presented in this report were developed using **Mermaid** and **PlantUML**, two text-based diagramming tools that allow diagrams to be created using simple textual syntax. The required diagram structures were first designed using Mermaid and PlantUML code according to the requirements of the Inventory Management System.

Mermaid was used for diagrams such as Data Flow Diagrams (DFD), Physical Entity Relationship Diagrams (PERD), and other system modeling diagrams where appropriate. **PlantUML** was used for UML-based diagrams, including Use Case, Component, Deployment, etc.

After defining the structure, entities, relationships, and interactions in Mermaid and PlantUML syntax, the diagrams were rendered into graphical representations using compatible diagram editors and rendering tools. The generated diagrams were then exported as images and inserted into the final report. **Google Docs** was used to organize and format the report, including the text, diagrams, headings, captions, and other required sections.

QN.1 Write the DFD for Inventory Management System

Data Flow Diagram (DFD)

A Data Flow Diagram (DFD) is a graphical representation of the flow of data through an information system. It shows how data enters the system, how it is processed, where it is stored, and how it is delivered to users.

Components of DFD

- External Entity: Source or destination of data outside the system.
- Process: Activity that transforms input data into output data.
- Data Store: Place where data is stored.
- Data Flow: Movement of data between entities, processes, and data stores.

Level 0 DFD (Context Diagram)

The Level 0 DFD represents the entire Inventory Management System as a single process and shows its interaction with external entities such as Supplier, Manager, and Customer.

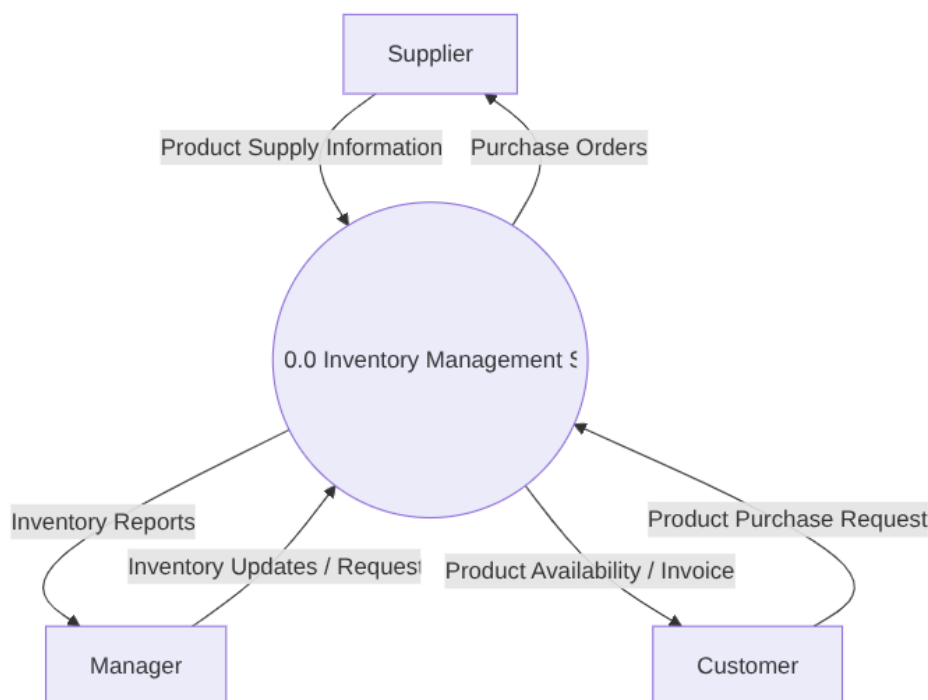


fig:Dfd-0 for Inventory Management System

Level 1 DFD

The Level 1 DFD decomposes the Inventory Management System into major processes such as Product Management, Stock Management, Order Management, and Report Generation along with their respective data stores.

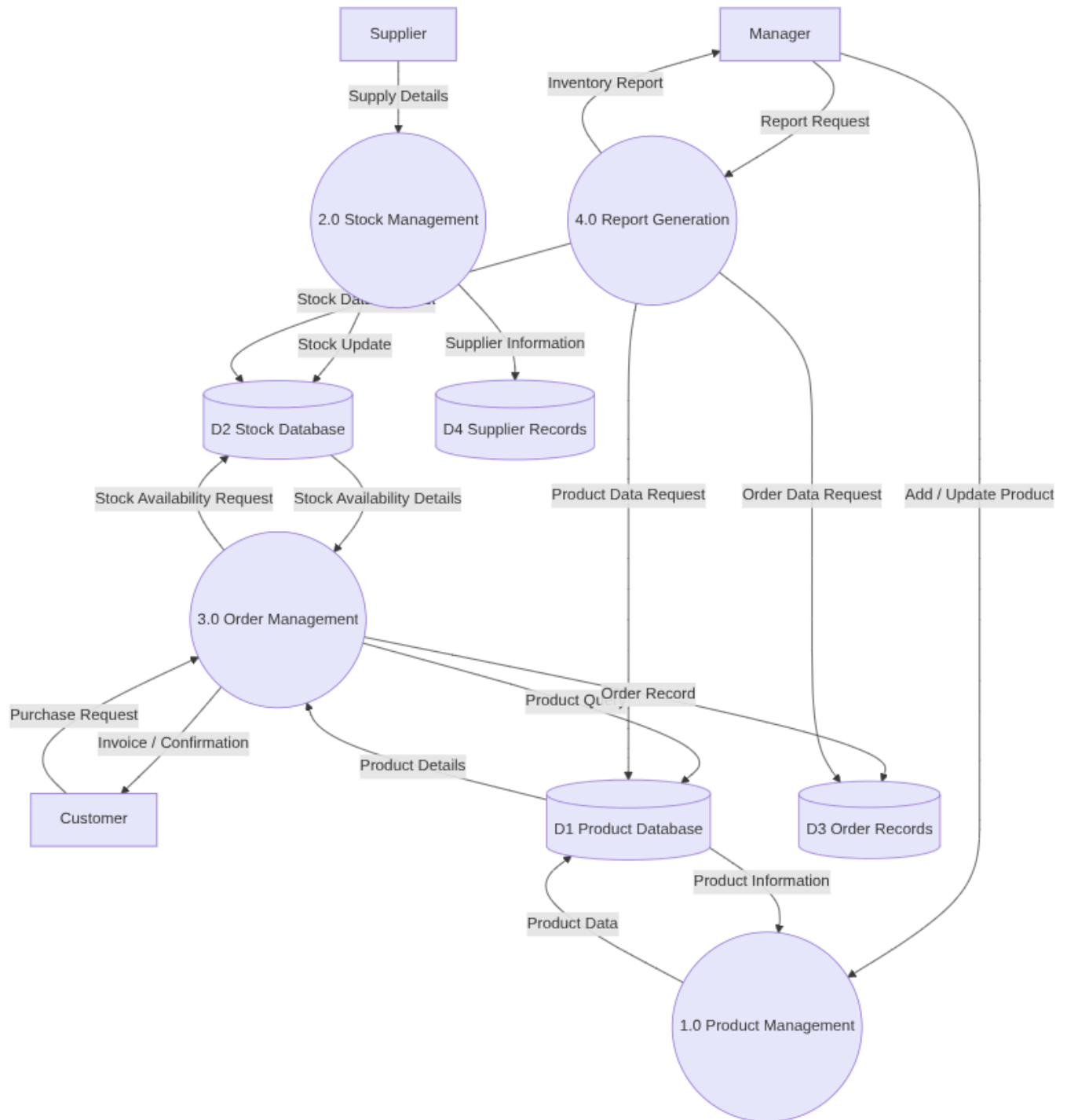


fig:Dfd-1 for Inventory Management System

Conclusion

The Level 0 DFD provides an overall view of the Inventory Management System and its interaction with external entities. The Level 1 DFD shows the internal processes, data stores, and data flow involved in managing products, stock, orders, and reports.

QN.2 Draw the Physical Entity Relationship (PER) Diagram for Inventory Management System

Physical Entity Relationship (PER) Diagram

A **Physical Entity Relationship (PER) Diagram** is a detailed representation of a database structure. It shows the entities, attributes, primary keys, foreign keys, data types, and relationships required for implementing the database.

Components of PER Diagram

- **Entity:** A database table that stores related information.
- **Attribute:** A column that represents a property of an entity.
- **Primary Key (PK):** An attribute that uniquely identifies each record in an entity.
- **Foreign Key (FK):** An attribute that references another entity to establish a relationship.
- **Relationship:** An association between two or more entities.
- **Data Type:** Specifies the type of data stored in an attribute.

PER Diagram for Inventory Management System

The Inventory Management System consists of the entities **Product**, **Inventory**, **Supplier**, **Customer**, and **Order**.

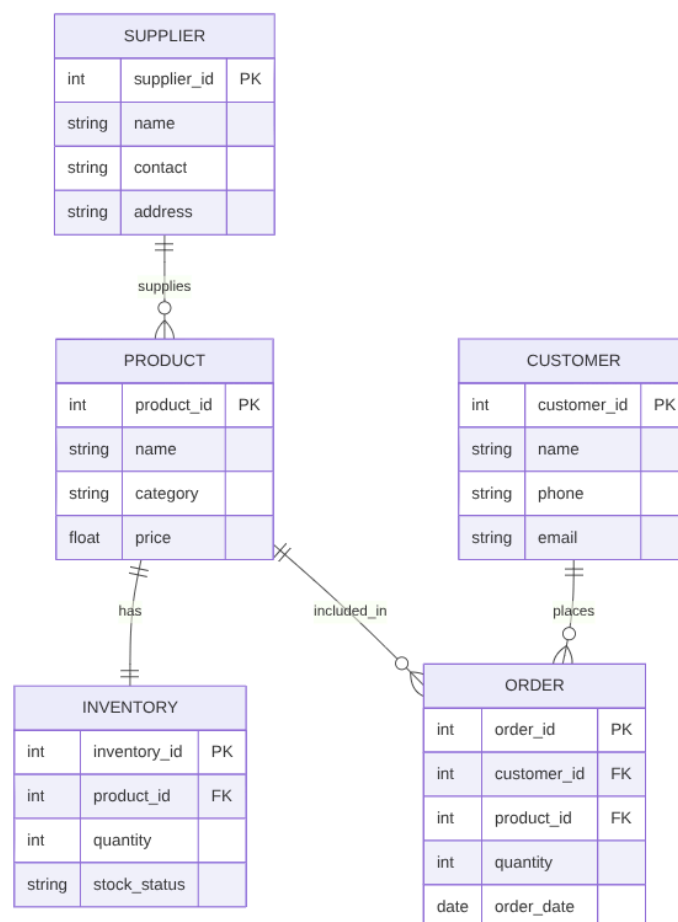


fig:PER Diagram for Inventory Management System

Relationship Description

- A Product has a corresponding Inventory record that stores stock information.
- A Supplier can supply multiple Products.
- A Customer can place multiple Orders.
- A Product can be included in multiple Orders.

Conclusion

The PER Diagram provides a detailed view of the Inventory Management System database by defining its entities, attributes, keys, data types, and relationships. It serves as the foundation for implementing and maintaining the database structure of the system.

QN.3 Draw the Use Case Diagram for Inventory Management System

Use Case Diagram

A Use Case Diagram represents the functional requirements of a system by showing the interactions between users (actors) and the functions (use cases) provided by the system.

Components of Use Case Diagram

- Actor: A user or external entity that interacts with the system.
- Use Case: A function or service provided by the system.
- System Boundary: Defines the scope of the system.
- Association: Shows interaction between an actor and a use case.

Use Case Diagram for Inventory Management System

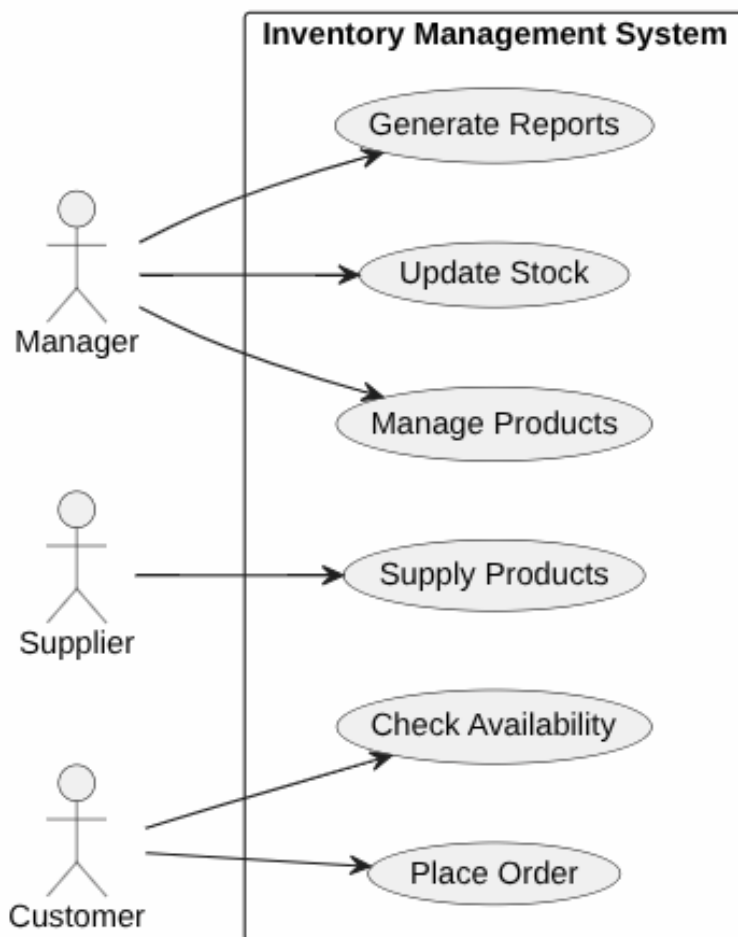


fig: UseCase Diagram for Inventory Management System

Description

- The Manager manages products, updates stock information, and generates inventory reports.
- The Supplier supplies products to the inventory system.
- The Customer checks product availability and places orders for available products.
- The Inventory Management System provides these functions and manages the interactions between the actors.

Conclusion

The Use Case Diagram represents the functional interaction between the Manager, Supplier, Customer, and Inventory Management System, providing a clear overview of the system's major functions.

QN.4 Draw the Class Diagram for Inventory Management System

Class Diagram

A Class Diagram is a UML structural diagram that represents the classes, attributes, methods, and relationships of a system. It provides a blueprint of the system's structure.

Components of Class Diagram

- Class: Represents an object or entity in the system.
- Attribute: Represents the data or properties of a class.
- Method: Represents the operations performed by a class.
- Association: Represents a relationship between classes.
- Multiplicity: Specifies the number of objects involved in a relationship.

Class Diagram for Inventory Management System

The main classes include Product, Supplier, Customer, Order, and Inventory. These classes work together to manage products, stock, suppliers, and customer orders.

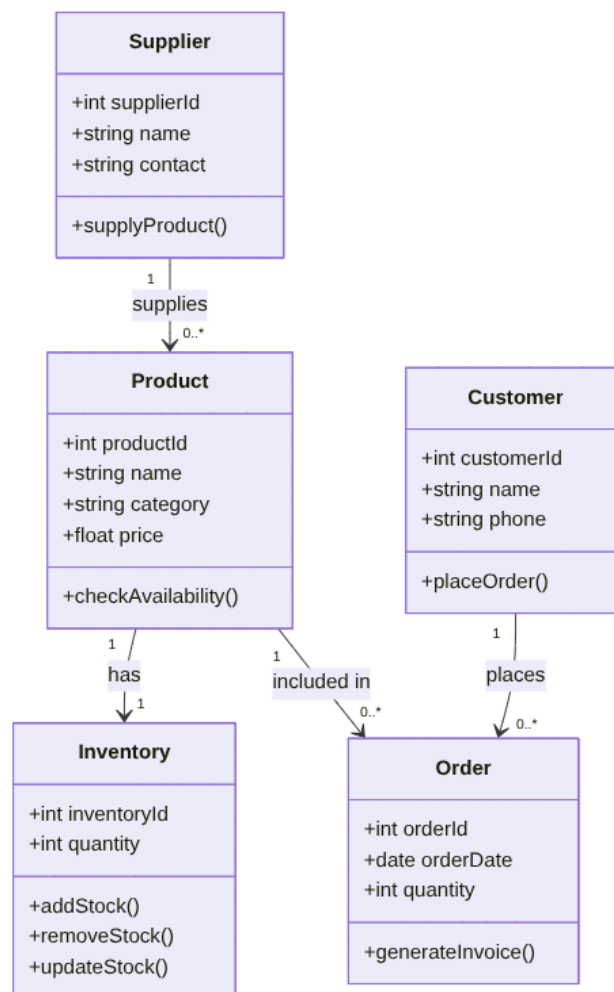


fig:Class Diagram for Inventory Management System

Class Description

- Product stores information about products available in the inventory.
- Inventory maintains the quantity and stock status of products.
- Supplier provides products to the inventory.
- Customers place orders for products.
- Order records customer purchase information and generates invoices.

Conclusion

The Class Diagram represents the static structure of the Inventory Management System by showing its main classes, attributes, methods, and relationships.

QN.5 Draw the Object Diagram for Inventory Management System

Object Diagram

An Object Diagram represents a snapshot of objects and their relationships at a particular point in time. It shows actual instances of classes with their attribute values.

Components of Object Diagram

- Object: An instance of a class.
- Attribute Values: Actual values assigned to an object's attributes.
- Link: Relationship between two objects.

Object Diagram for Inventory Management System

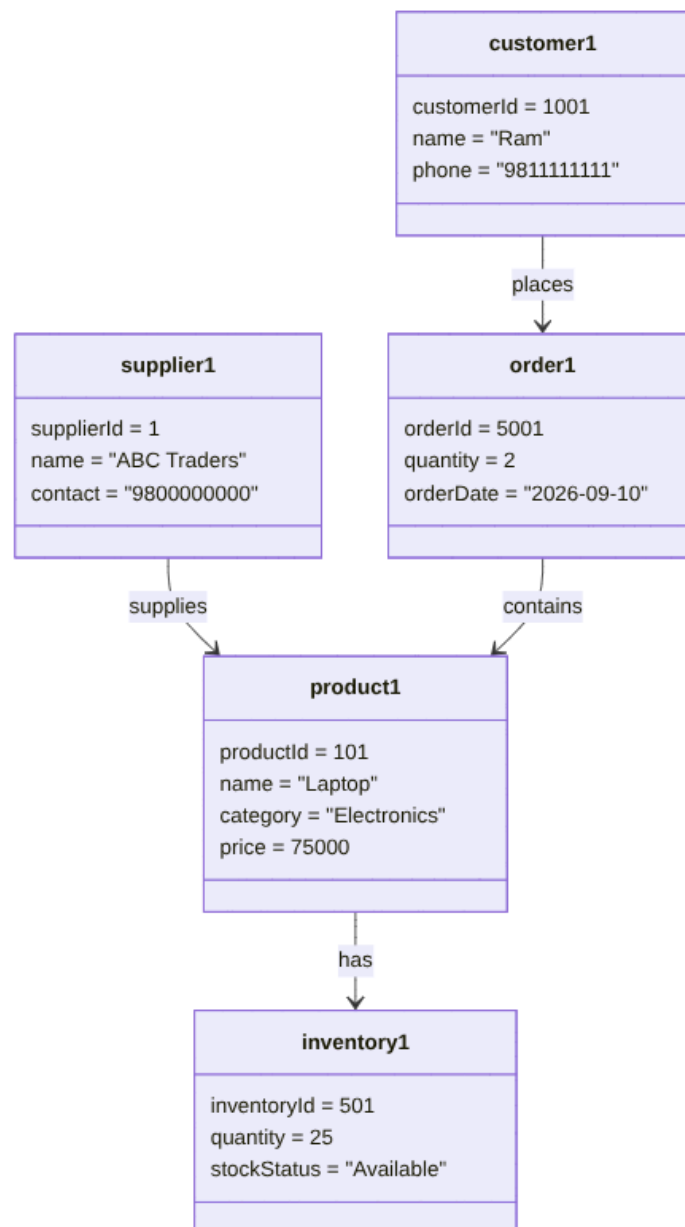


fig: Object Diagram for Inventory Management System

Description

- The product1 object represents a specific product, Laptop, with its product ID, category, and price.
- The inventory1 object represents the inventory record of the laptop and stores its current quantity and stock status.
- The supplier1 object represents a specific supplier who supplies the laptop product.
- The customer1 object represents a customer named Ram who places an order.
- The order1 object represents a specific order containing two units of the laptop.

Conclusion

The Object Diagram shows a real-time snapshot of specific objects in the Inventory Management System and illustrates how these objects are related to each other.

QN.6 Draw the Component Diagram for Inventory Management System

Component Diagram

A Component Diagram is a UML structural diagram that shows the major software components of a system and the dependencies between them. It represents how different modules work together to provide system functionality.

Components of Component Diagram

- Component: A modular part of the software system.
- Interface: Defines how components communicate with each other.
- Dependency: Shows that one component depends on another.
- Connection: Represents communication between components.

Component Diagram for Inventory Management System

The Inventory Management System consists of components such as User Interface, Product Management, Stock Management, Order Management, Report Management, and Database.

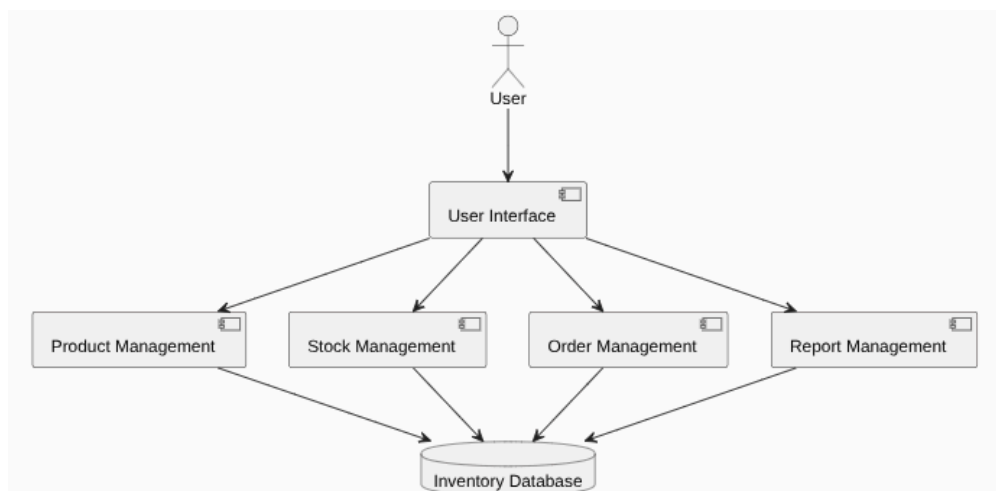


fig:Component Diagram for Inventory Management System

Component Description

The User Interface allows users to interact with the Inventory Management System. Product Management handles product information, Stock Management manages inventory levels and stock updates, Order Management processes customer orders and invoices, and Report Management generates inventory and sales reports. All modules store and retrieve data from the Inventory Database.

Conclusion

The Component Diagram shows the major software components of the Inventory Management System and their dependencies. It provides a simple view of how different modules communicate with the database to perform inventory operations.

QN.7 Draw the Deployment Diagram for Inventory Management System

Deployment Diagram

A Deployment Diagram is a UML diagram that represents the physical deployment of software components on hardware devices or execution environments. It shows how different nodes communicate with each other.

Components of Deployment Diagram

- Node: A physical device or execution environment.
- Artifact: Software deployed on a node.
- Communication Path: Connection between different nodes.
- Deployment: Represents software deployed on a particular node.

Deployment Diagram for Inventory Management System

The Inventory Management System can be deployed using client devices, an application server, and a database server. Users access the application through client devices, while the application server communicates with the database server.

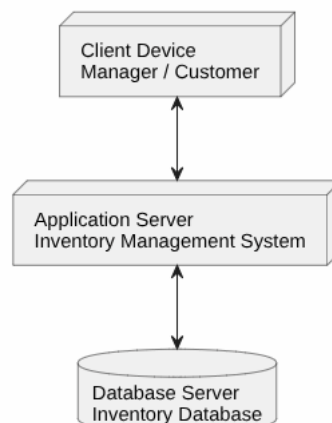


fig:Deployment Diagram for Inventory Management System

Deployment Description

The Client Device is used by managers and customers to access the Inventory Management System. The Application Server runs the main inventory application and handles system operations. The Database Server stores product, stock, supplier, and order information.

Conclusion

The Deployment Diagram shows the physical architecture of the Inventory Management System and illustrates how client devices, the application server, and database server communicate with each other.

QN.8 Draw the Activity Diagram for Inventory Management System

Activity Diagram

An Activity Diagram is a UML behavioral diagram that represents the workflow of activities in a system. It shows the sequence of actions and decision points from the beginning to the end of a process.

Components of Activity Diagram

- Initial Node: Starting point of the activity.
- Activity: A task or operation performed by the system or user.
- Decision Node: Represents a condition where the workflow branches.
- Control Flow: Shows the direction of the workflow.
- Final Node: Represents the end of the activity.

Activity Diagram for Product Purchase

The following activity diagram represents the process of purchasing a product in the Inventory Management System.

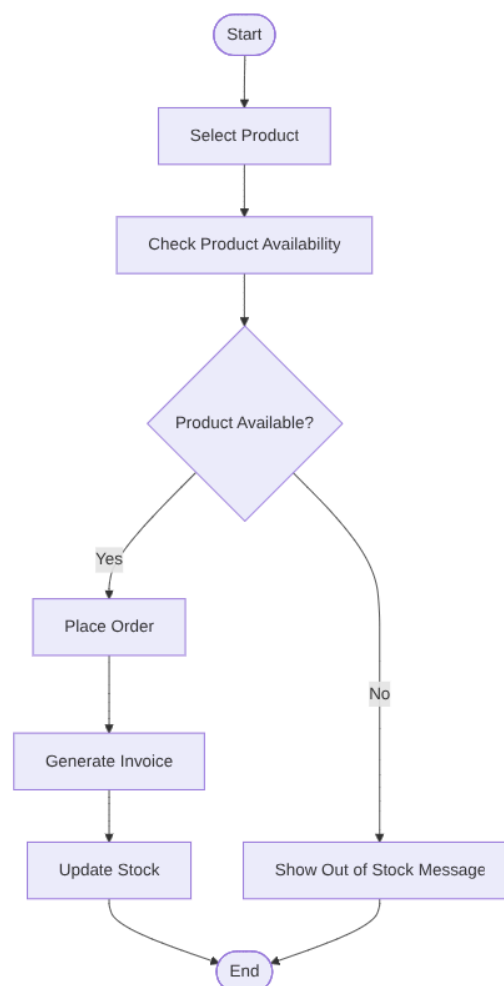


fig:Activity Diagram for Inventory Management System

Activity Description

The customer selects a product and the system checks its availability. If the product is available, the order is placed, an invoice is generated, and the stock is updated. If the product is unavailable, an out-of-stock message is displayed.

Conclusion

The Activity Diagram shows the workflow of the product purchase process and clearly represents the activities and decisions involved in inventory management.

QN.9 Draw the Sequence Diagram for Inventory Management System

Sequence Diagram

A Sequence Diagram represents the interaction between objects or components of a system in a particular sequence over time. It shows how messages are exchanged between participants to perform a specific operation.

Components of Sequence Diagram

- Actor: External user who interacts with the system.
- Participant: Object or system component involved in the interaction.
- Message: Communication between participants.
- Sequence: The order in which interactions occur.

Sequence Diagram (Product Purchase Subset)

Subset Used: *Product Purchase Process*

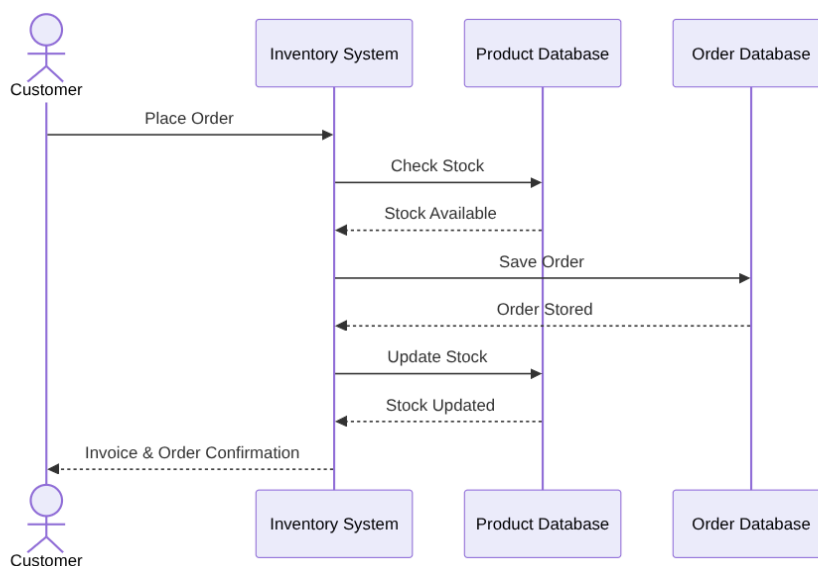


fig:Sequence Diagram for Inventory Management System

Description

The Customer starts the process by placing an order through the Inventory Management System. The Inventory System checks the availability of the requested product from the Product Database. If the product is available, the system saves the order in the Order Database. After successfully saving the order, the system updates the product stock quantity. Finally, the system sends the invoice and order confirmation to the customer.

Conclusion

The Sequence Diagram shows the step-by-step interaction between the customer, inventory system, and databases during the product purchase process.